Full-cache queries: second large matched recovery

Run fullcache-20261101-v3-funded · RTX 4090 · 2026-11-01 UTC

The second original 63-bit public input (seed202) completed all three recovery executions on one fresh RTX 4090. Every certificate passed fresh CPU replay and scalar+1 rejection; certificates and relation/attempt counts matched across arms. The candidate took 1053.897 seconds, compared with 1183.824 and 1185.764 seconds for the identical baseline executable before and after it. Candidate divided by baseline mean was 0.889519. The observed cold-time reduction was 11.05%. Baseline repeat spread was 0.164 percent.

Run	Arm	Order	Cold solver s	Outcome
63-202-0-x	Baseline repeat	First	1183.824	Verified
63-202-0-v	Full-cache candidate	Second	1053.897	Verified
63-202-0-b	Baseline	Third	1185.764	Verified

The plot contains the three new executions on pod k1mjc0sf1gpzvq. Cold solver clocks include target construction, startup, curve validation, base construction, relation queries, elimination, in-process replay, serialization and shutdown. Additional fresh CPU replay is recorded separately. One round for one fixed input; shared-host L0 descriptive clocks. No profiler was attached.

Checks, history and reproducibility

Frozen setup and checks

Version 3 was frozen before new data. It selects the original seed202 input in x/v/b order and retains the exact GPU/driver code, resource limits, checks and analysis formulas. The original worker was externally deleted; all three arms were restarted on this host. No old-host clock is substituted or pooled. The query specialization directly accesses the complete cached right table and reuses target conversion. Original normalization, lookup and minimum-winner semantics remain unchanged.

Each new recovery used 344 relations and 13,733 attempts. All 29 driver tests and Clippy passed. Host and device arithmetic each passed 37,584 cases. The direct query differential passed 315 cases, and the backend check covered full-cache refusal and fallback. Six sanitizer executions reported zero errors/hazards. Service/driver self-tests and the PTX check for unwanted profiling instructions passed before the recovery runs.

The original public fixture was regenerated and compared byte-for-byte. Each successful recovery required a full right cache and nonempty filter. Tables had 549,596,520 left rows and 444,762,500 cached right rows. Conservative admission was 14,920,341,256 bytes; this is an admission estimate, not measured peak memory.

Preserved history

Version 1 failed preflight before any recovery execution. Version 2 completed 30 small recoveries and the three-arm first large comparison, then lost its pod during the second input's baseline. That interrupted execution has no complete clock; two further version 2 slots were never started. The first version 3 submission returned HTTP 402 before allocation and consumed no GPU time.

The new version 3 set has three planned, attempted and verified executions. Together with version 2 there are 36 verified recoveries and one interrupted execution across 37 actual recovery attempts. The two unstarted version 2 slots remain recorded. Both original large inputs now have complete within-host matched comparisons, on different hosts. The original whole-panel analyzer and its incomplete-version 2 refusal remain unchanged.

Locate and reproduce

Protocol: `protocol-v3.json`, hash `ebffb8127816d9b5b58785cc391b6a7cf3882718bbb5c0bc33e8e92c6f60d34a`. Instrument: `64901ebb2182790f0629ed637b507aea4da48bee`, clean source. Command: `bash research/explicit_relations_fullcache_20261101/run-v3.sh`. Source archive SHA256: `3424947660018b11d1b6bb1d601ad671b79b25f1aa29ccdbb2db95ae59273713`.

Results and canonical export: `results-v3/`. Analysis source: `analyze-v3.rs`. Hashes, raw archive location and every producer file are in `results-v3/archive-manifest.json`. Hydrate large logs with `python3 tools/artifacts/artifacts.py pull research/explicit_relations_fullcache_20261101/results-v3`.

The single worker was bounded at 9,000 seconds and the controller at 9,300 seconds, within the unused original budget. Request through verified cleanup took 3,780.073 seconds. The conservative charge was 3,840 seconds, leaving 5,160 seconds of the original worker budget. No GPU remains running. Budget and verified deletion receipts are retained with the final results. This report is rendered only after successful collection, replay and export checks, and after raw records are published.